

MORELIA, Mexico

WMO: 766650

Lat: 19.7217N

Lon: 101.1831W

Elev: 1913

StdP: 80.36

Time Zone: -6.00 (MEX)

Period: 83-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.3	7.4	-1.6	4.2	20.5	0.1	4.8	18.6	8.4	18.0	7.1	21.6	0.9	210	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.4	31.6	16.6	30.3	16.7	29.1	16.4	20.3	26.0	19.6	25.5	19.0	25.0	2.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	17.0	23.7	17.7	16.1	22.7	17.0	15.4	21.9	67.8	25.7	65.0	25.7	62.5	24.9	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	4.8	3.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	15.5	17.2	18.6	21.2	21.9	21.1	19.5	20.0	19.7	18.8	17.0	16.1	(6)
(7)		DBStd	2.64	1.67	1.90	2.25	1.79	1.81	1.90	1.11	1.10	1.15	1.71	1.96	1.80	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	297	88	41	22	2	1	1	3	2	3	15	48	72	(9)
(10)		CDD10.0	3246	171	201	268	337	370	332	295	309	292	274	209	188	(10)
(11)		CDD18.3	500	0	8	31	89	112	83	40	52	45	31	7	2	(11)
(12)		CDH23.3	4459	58	213	474	1059	1082	570	196	255	174	201	109	67	(12)
(13)	CDH26.7	1027	1	21	88	376	352	147	9	9	4	12	5	2	(13)	
(14)	Wind	WSAvg	1.5	1.6	1.7	1.7	1.6	1.4	1.5	1.3	1.3	1.4	1.5	1.4	1.4	(14)
(15)	Precipitation	PrecAvg	970	17	16	14	11	52	162	217	197	176	76	18	15	(15)
(16)		PrecMax	1240	166	182	170	40	133	254	416	327	269	149	61	222	(16)
(17)		PrecMin	658	0	0	0	0	0	61	82	119	95	20	3	0	(17)
(18)		PrecStd	153	32	35	32	12	35	49	67	53	45	35	17	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	28.6	30.8	33.0	33.4	31.6	27.8	27.8	27.3	28.1	27.4	26.7	(19)
MCWB			14.7	15.4	16.9	16.7	17.2	17.9	18.1	18.3	18.9	17.7	16.6	16.0	(20)	
2%		DB	24.8	27.1	28.9	31.6	31.7	29.8	26.4	26.7	26.2	26.6	25.9	25.1	(21)	
		MCWB	14.2	15.0	16.5	16.2	16.6	17.9	18.2	18.3	18.2	17.2	16.2	15.6	(22)	
5%		DB	23.6	25.9	27.5	30.3	30.3	28.4	25.4	25.9	25.2	25.5	24.6	23.7	(23)	
		MCWB	14.1	14.6	15.8	16.0	16.8	17.8	18.1	18.1	17.9	16.8	15.8	14.9	(24)	
10%		DB	22.3	24.5	26.2	29.0	28.9	26.9	24.3	25.0	24.3	24.4	23.3	22.4	(25)	
		MCWB	13.5	14.0	15.1	15.9	16.2	17.7	17.9	18.0	17.7	16.5	15.5	14.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	17.5	19.6	19.2	20.2	21.6	20.6	20.8	20.7	20.0	18.9	17.9	(27)
MCD B			23.2	26.2	27.6	29.9	28.6	27.6	25.1	25.5	24.9	25.4	24.4	23.3	(28)	
2%		WB	15.2	16.2	18.2	18.0	19.1	20.0	19.5	19.8	19.6	18.9	17.9	16.9	(29)	
		MCD B	23.1	25.0	26.6	29.1	27.3	26.5	24.5	25.0	24.5	24.4	23.3	22.8	(30)	
5%		WB	14.5	15.3	17.0	17.3	18.4	18.9	18.7	18.9	18.8	18.1	17.0	16.0	(31)	
		MCD B	22.4	24.2	25.8	28.2	26.7	25.1	23.9	24.3	23.8	23.6	22.5	22.2	(32)	
10%		WB	13.8	14.5	15.8	16.4	17.7	18.2	18.1	18.2	18.1	17.4	16.2	15.1	(33)	
		MCD B	21.4	23.0	24.8	27.0	25.9	24.5	23.4	23.7	23.0	22.6	21.9	21.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.9	15.5	15.8	16.9	14.4	11.2	10.4	10.6	10.2	12.3	14.1	14.5	(35)
MCDBR			16.9	17.6	17.7	18.6	16.6	13.6	12.1	11.9	11.6	13.9	15.6	16.2	(36)	
MCWBR			7.7	7.3	7.6	7.0	6.0	4.8	5.3	4.9	4.9	5.8	7.1	7.8	(37)	
5% WB		MCD BR	15.8	16.4	16.6	17.2	14.1	11.9	11.0	11.3	10.8	12.3	13.9	15.4	(38)	
		MCWBR	7.7	7.2	8.2	7.3	5.9	5.3	5.8	5.5	5.3	5.8	7.1	8.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.331	0.340	0.334	0.377	0.460	0.419	0.404	0.401	0.404	0.373	0.331	0.328	(40)
taud		2.427	2.393	2.406	2.302	2.096	2.247	2.300	2.306	2.303	2.389	2.463	2.462	(41)		
Ebn at Noon		948	957	973	926	843	870	884	893	891	911	942	941	(42)		
Edh at Noon		107	117	120	135	164	140	133	133	131	116	102	101	(43)		
(44)	All-Sky Solar Radiation	RadAvg		4.85	5.76	6.67	7.14	6.58	5.76	5.61	5.62	5.08	5.15	5.00	4.74	(44)
(45)		RadStd		0.36	0.53	0.37	0.26	0.40	0.38	0.35	0.31	0.32	0.34	0.38	0.27	(45)

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (5 neighbors)	N/A	+0.80	-1.73	N/A	-0.74	-0.94	-30	N/A	N/A	N/A	(47)		

Nomenclature: See separate page